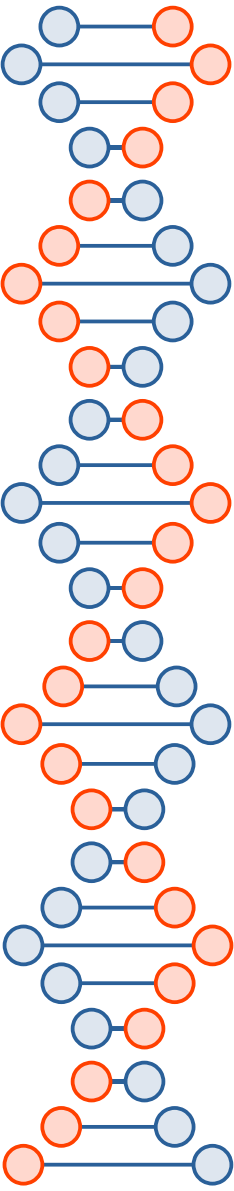


GitHub

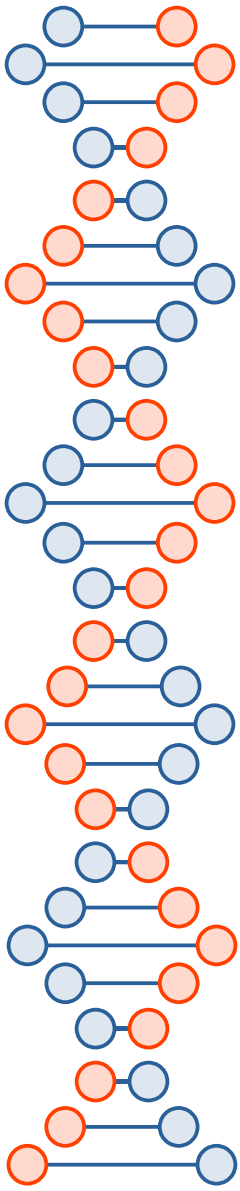
Presentation to T-Exchange
October 2026

Norman Dunbar



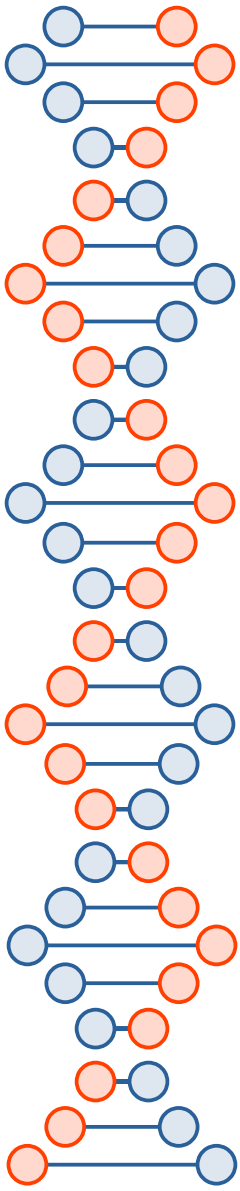
Introduction – What's Coming

- What is Git?
- What is GitHub & its Alternatives?
- Very Simple GitHub
- Simple GitHub
- Medium GitHub
- Slightly Advanced GitHub
- Advanced GitHub



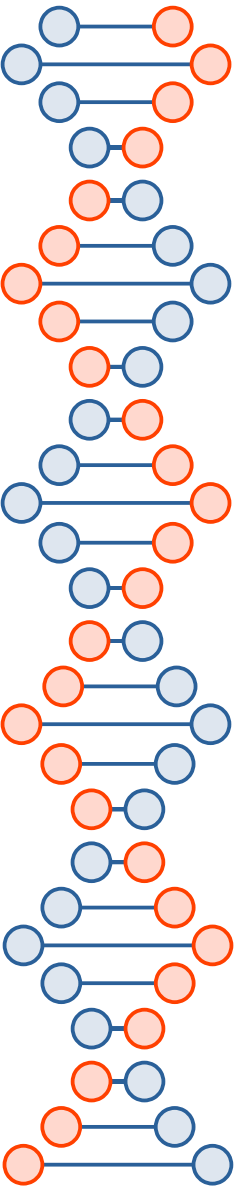
What is Git

- Git is a version control system---Linus Torvalds wrote it
- It protects your code from disasters
- It is distributed---everyone has a full copy and all history
- New changes are tested in branches
- Branches are merged when complete
- Free book at <https://git-scm.com/book/en/v2>
- Requires a Git client



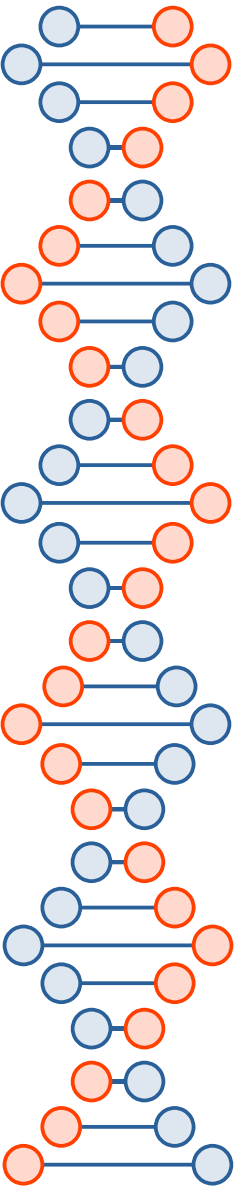
What is GitHub?

- Online code repository of millions of projects
- Uses Git
- Provides off-site backups for projects
- Secure, in as much as being owned by Microsoft can be!
- Probably compromised by Microsoft's "AI" (LLM!)
- Private and Public repositories allowed
- Free or pay-for-use.



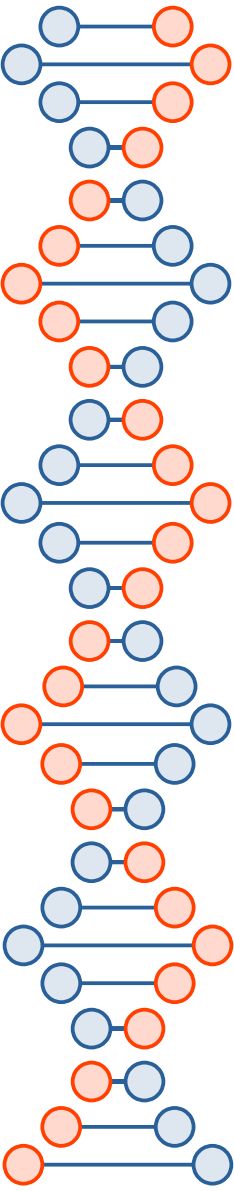
Alternatives to GitHub

- GitLab at <https://gitlab.com>
 - Private & Public repositories
 - Not Microsoft!
 - Similar to GitHub
 - My books live on GitLab



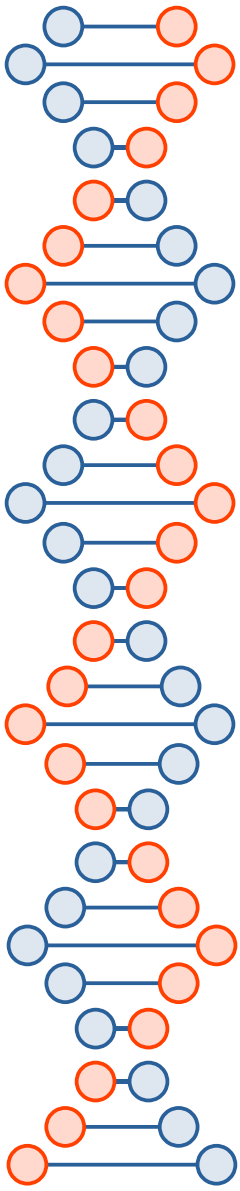
Alternatives to GitHub

- CodeBerg at <https://codeberg.org>
 - Public repositories only
 - Free and Open Source projects only
 - Web pages offered
 - Not Microsoft!
 - “AI” is not permitted
 - I have an account, but unused as yet



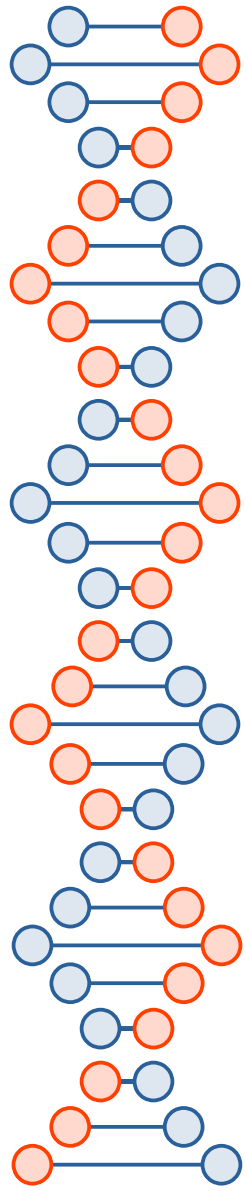
Very Simple GitHub

- No account required
- You only want to run the code, read the book etc
- You will not be changing anything.
- You download a project's release
- You *can* download the code too, but don't have to.



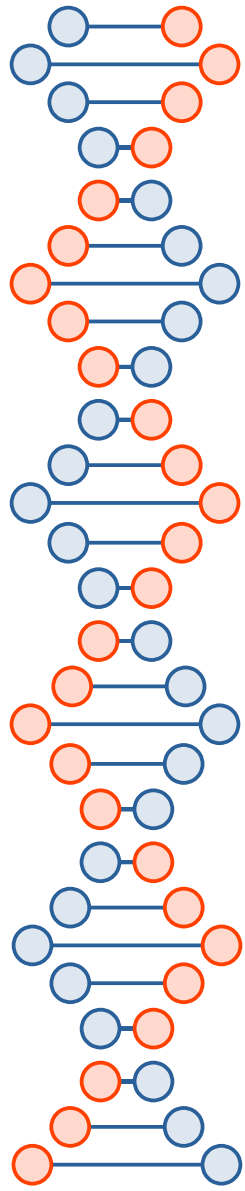
Very Simple GitHub - Downloading

- Click on the GitHub URL you have been given
- Click on “releases”, if you are at the repository home page
- Find the desired release---there may be more than one
- Open the Assets list
- Click to download all required assets
- Optionally, ignore the two “Source Code” assets
- Enjoy your download!



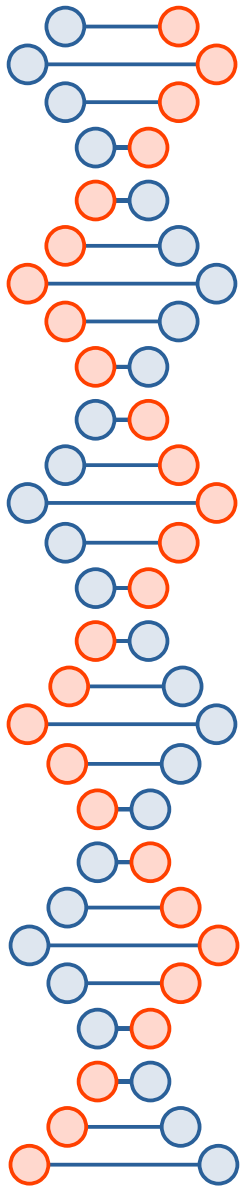
Very Simple GitHub - Searching

- Searching may be required to find a user's repositories
- Click "Search"
- Type, eg "Norman Dunbar" with quotes, press return
- Filter by "users" (left side of page)
- Click "NormanDunbar"
- You are now at Norman's home page
- You need to find a repository with "Assembly" in the name, so...



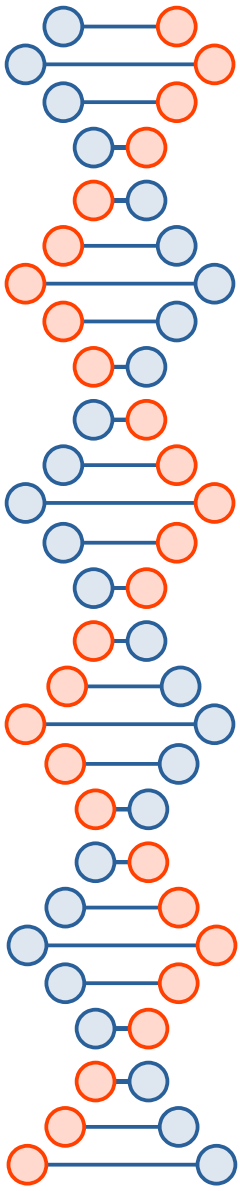
Very Simple GitHub - Searching

- Click “Search”
- Leave “owner:NormanDunbar” and add “Assembly”
- Press return
- Click on the most suitable filename on the results page
- On the next page, click the repository name (QLAssemblyLanguageMagazine) to go to the repo.
- Proceed as previously instructed to get the code release.
- If nothing was found, then...



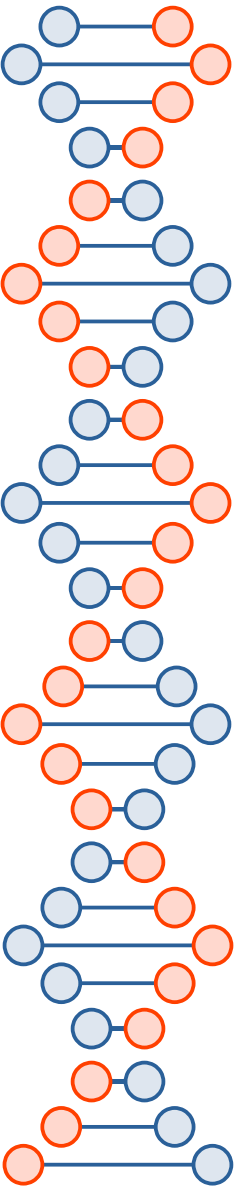
Very Simple GitHub - Searching

- We didn't get a search hit, go back to NormanDunbar's home page
- Click "Repositories" at the top left
- Click "Sort" at the top right, select "Name"
- Scroll down, looking for the repo you need
- Click "Next >" at the bottom of each page to continue
- Found it? Proceed as previously to get hold of the code
- Not found? Bugger!



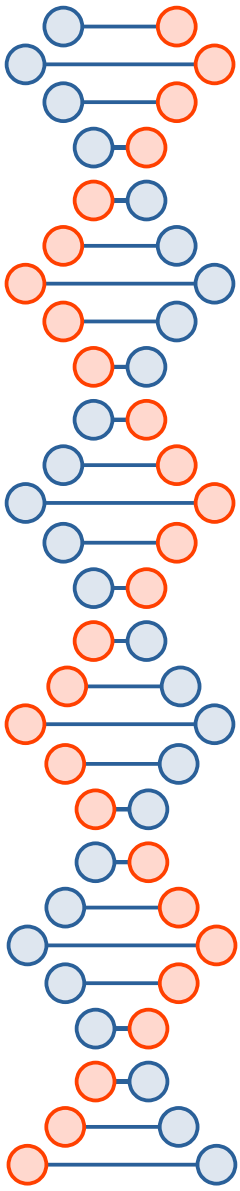
Simple GitHub

- No account necessary
- No login required
- We can download the source code, as a Zip file, for our own use
- We have no intention to make changes to the source code---but we could



Simple GitHub – Zip Downloads

- Connect to <https://github.com> or to the URL for the repository---for example
<https://github.com/NormanDunbar/QLAssemblyLanguageMagazine>
- Click 
- Click “Download ZIP”
- Unzip and use the source code
- You *might* get an executable, if not, check the releases

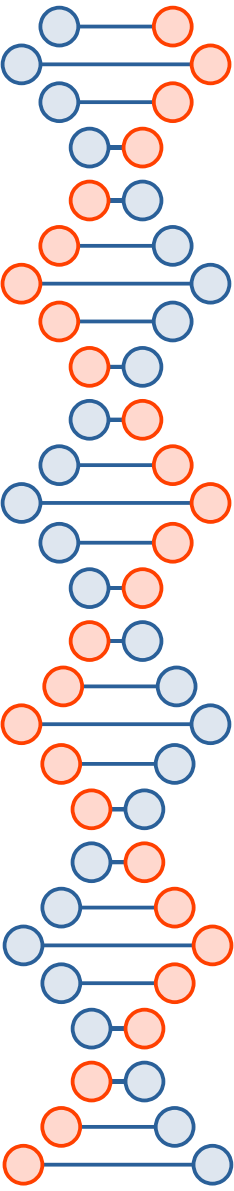


Medium GitHub

- No Account required
- No login required
- We can download source code for our own use
- We may intend to make changes, but only for ourselves
 - Download Zip file as per Simple GitHub, then amend the source code---keeps it simple!
 - “Clone” the repository locally, then change it---better!
- We have no intention of sharing our changes with the original authors

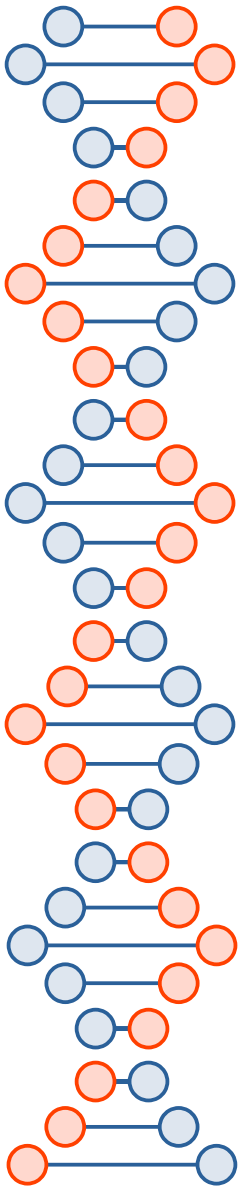
Medium GitHub - Cloning

- The simplest option is to proceed as per Very Simple or Simple GitHub
- Connect your browser to the required repository
- Click 
- Click “HTTPS”
- Click the copy icon---two sheets of paper---to copy the repository URL
- Then...



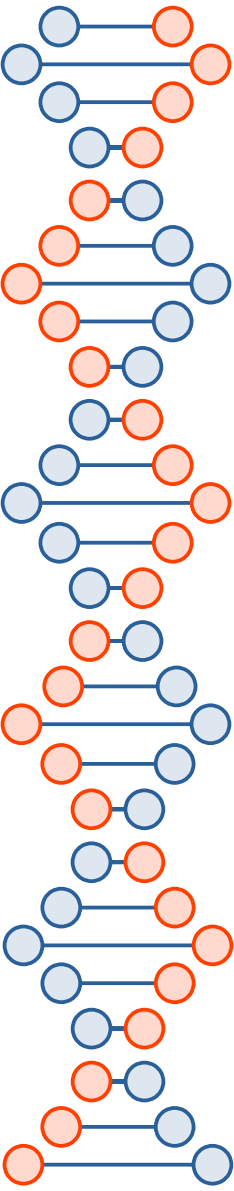
Medium GitHub - Cloning

- In a terminal session, navigate to where you normally store source code. For me, it's "SourceCode"
- Type "git clone " and paste the repository URL
- `Git clone
https://github.com/NormanDunbar/QLAssemblyLanguageMagazine.git`
- Press return
- Wait...
- You now have a clone of the repository and the full history



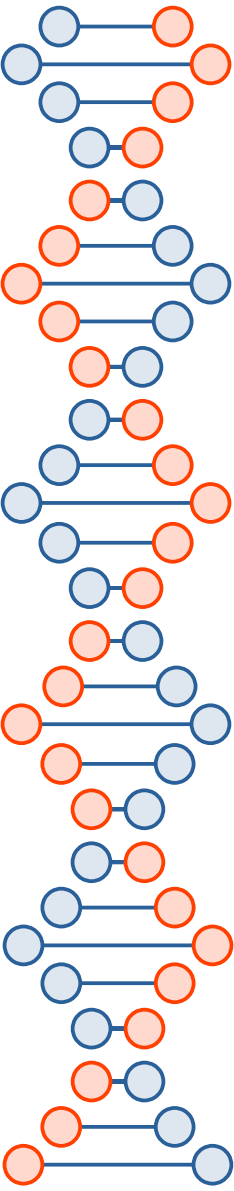
Medium GitHub – Updating Clones

- If you have not made any important changes to the code, you can update it to the authors' latest versions any time you feel. Simply:
 - Open a terminal session
 - Change to `SourceCode/QLAssemblyLanguageMagazine` (to continue with our example)
 - Run the command `git pull`
 - Any changes you made have a strong possibility of being overwritten and lost. Best to avoid this if you need to keep your changes



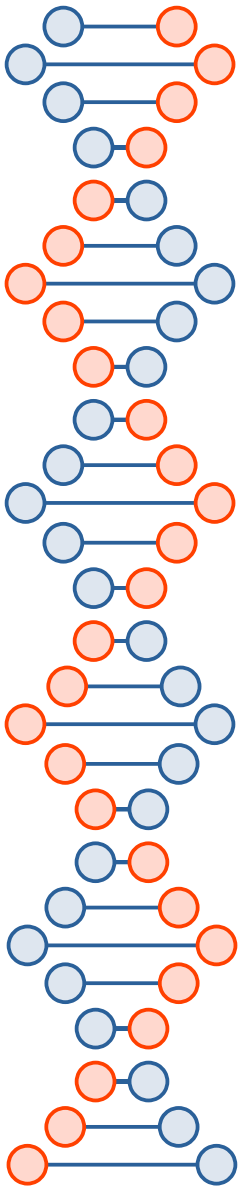
Slightly Advanced GitHub

- Account required
- Login required
- We will download source code for our own use
- We intend to make changes to the source code
- We will use the code for ourselves, but may also share our changes with the original authors
- We therefore need to “fork” the original repository



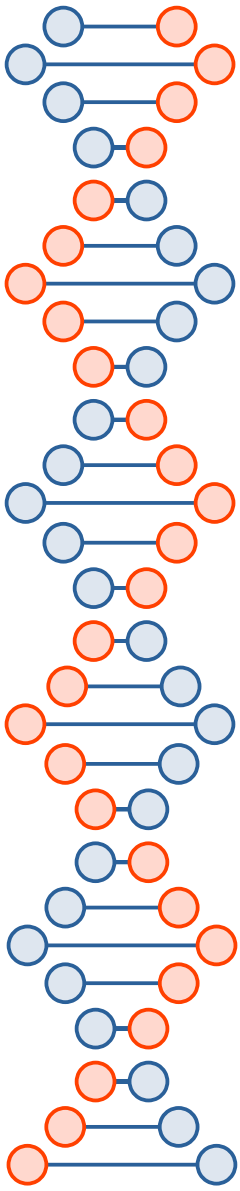
Slightly Advanced GitHub - Forking

- Stop sniggering at the back!
- Login to <https://github.com>
- Open or search for the repository to be forked
- On its home page, click “Fork”
- Check suggested details, correct any as required
- Click “Create fork”
- You now have a fork of the repository
- Clone the fork, to your PC as already discussed



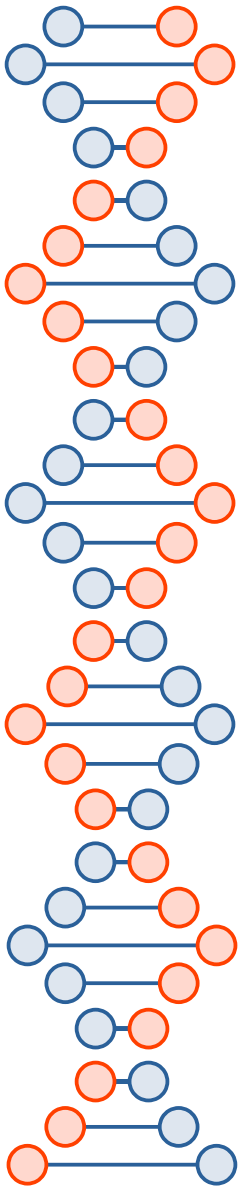
Slightly Advanced GitHub – Using the Clone

- Create a branch: `git checkout -B my_branch`
- Check our branch: `git status`
- Edit the code, compile, test, repeat!
- Find all our changes: `git status`
- Schedule changes:
 - `git add -all`
 - `git add <list of files>`
- Commit changes: `git commit -m "commit message here"`



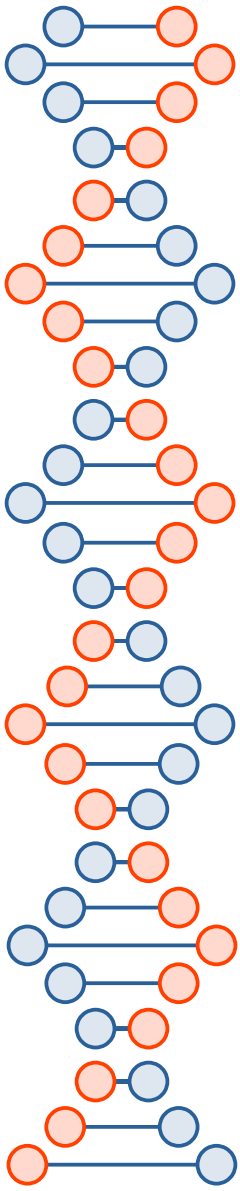
Slightly Advanced GitHub – Using the Clone

- Merge changes into “main” branch:
 - Checkout main: `git checkout main`
 - Merge from branch: `git merge my_branch`
- Push commits to fork: `git push`
- Optionally, tag the new state: `git tag -a -m "tag message" tag_name`
- Delete branch: `git branch -D my_branch`



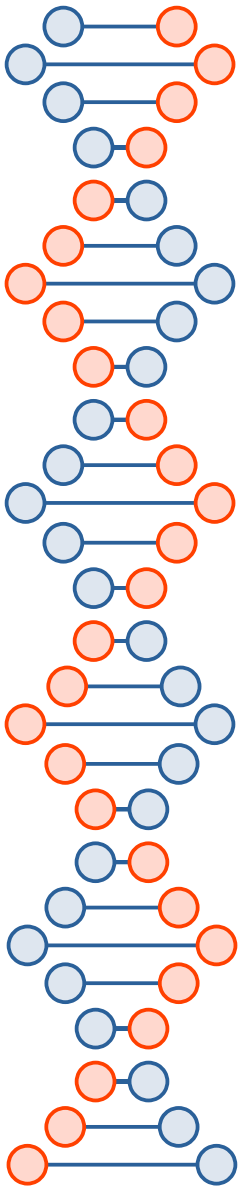
Slightly Advanced GitHub – Fork for Everyone

- Proceed as per Fork for Ourselves, however:
 - After pushing changes to our fork:
 - **DO NOT** delete the branch created for the changes!
 - Raise a merge request
 - Reference any “issues” it fixes
 - Wait for original authors to merge our changes
 - **Now** you can delete the branch
- We have now shared our changes



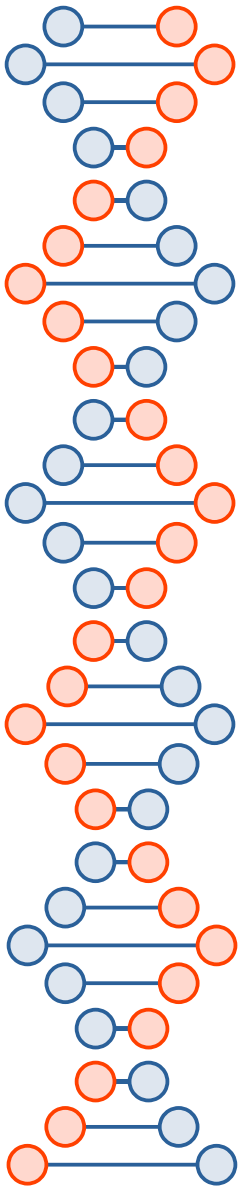
Slightly Advanced GitHub – Fork Updates

- Login to our fork's homepage on GitHub
- Check for a message “*This repository is nn commits ahead of, and nn commits behind <original repository>*”
- Click the “Sync Fork” button if the message is there and lists any “commits behind” the original repository
- That's it done, our changes are not affected or lost
- Need to update our clone also...



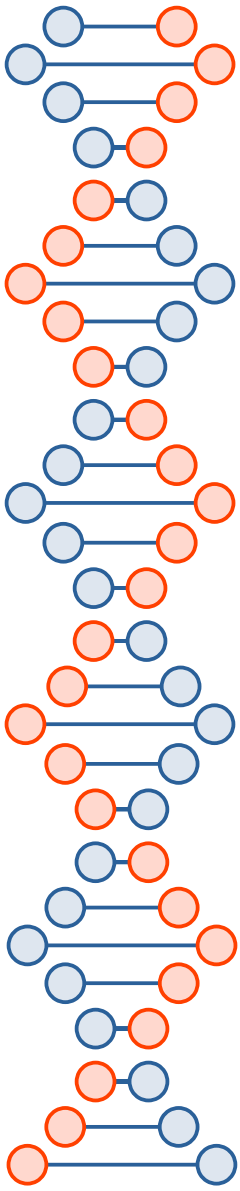
Slightly Advanced GitHub – Clone Updates

- Start a terminal session on our PC
- Navigate to where we keep the repository clone
- We need to be inside the cloned repository, anywhere inside is fine
- Execute “git pull”
- Our clone is now up to date with our fork



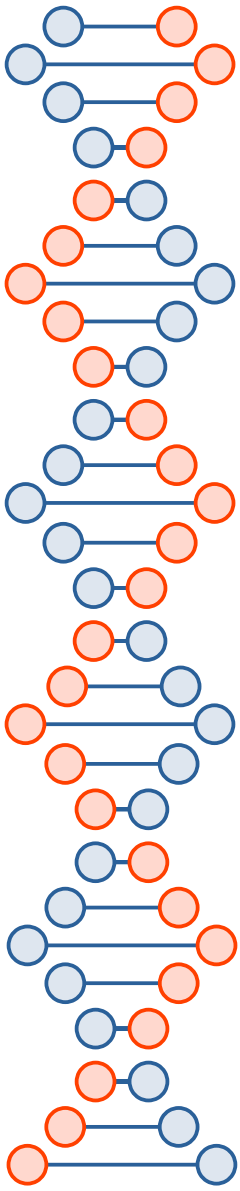
Slightly Advanced GitHub – Raise Merge Request

- Login to the fork's home page
- Click “Pull requests”
- Click “New pull request” button
- Check/fill in details
- Click “Create pull request”
- Fill in details, be meaningful and complete
- Click “Create pull request”
- ***DO NOT*** delete the branch!



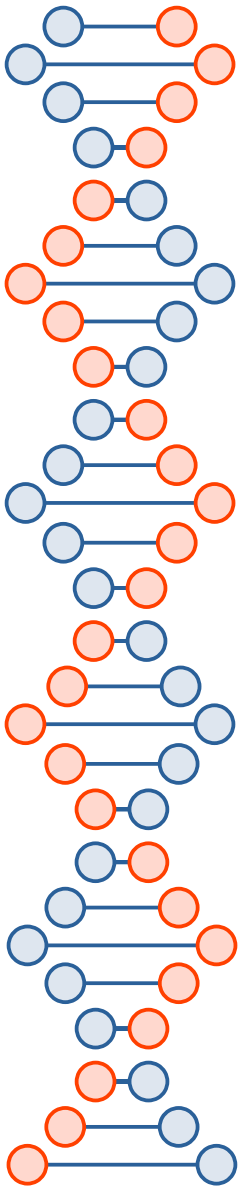
Slightly Advanced GitHub – Raise Merge Request

- Original repository developers will:
 - Get a notification about your pull request
 - Click “Pull requests” in the repository
 - Checks your details about the change
 - Might initiate a conversation for more details etc
 - Clicks “Merge pull request”
 - Updates the details, which you will receive back
 - Clicks “Confirm merge”



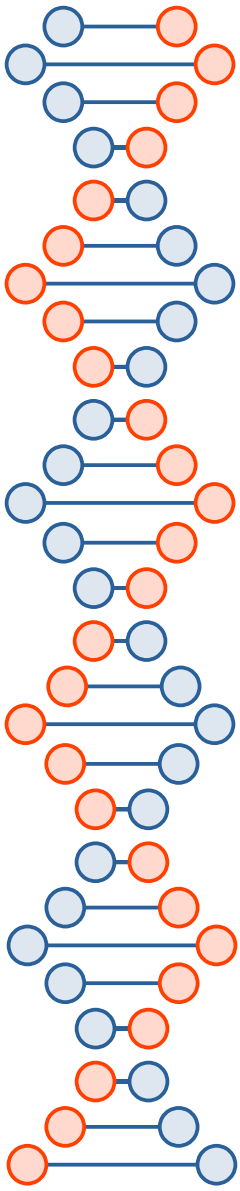
Slightly Advanced GitHub – Raise Merge Request

- You login to https://github.com/your_name
- You see a blue dot by the “in-tray” icon---you’ve got mail!
- Click the tray
- Click the notification about the pull request
- Will see “successfully merged and closed”---success!
- You can **now** delete the branch created for the change
- Sit back, knowing the world uses your code!



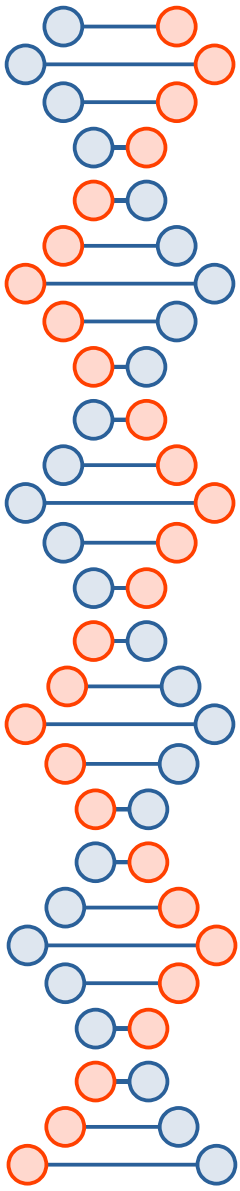
Advanced Github

- Account required
- Website creation - individual/organisation
- Website creation – projects
- Only permitted for public repositories
- Github Pages



Advanced Github

But all that is a subject for another day and/or another presentation.



The End

- Wake up at the back, I'm finished!
 - Any questions?
- Good, now, it's tea and cake time!
 - I'm off home with my dog!